

Pricing the Optionality the BuckBasket Sells

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We analyze the Panoptic carry/convexity critique against the BuckBasket and specify the wave-3 markout instrument. We evaluate the basket's own side of the excursion catalogue and record six design decisions reached by working each open question through one historical and one synthetic example.

We map the system as it stands across every design document in the repo, state the theory each decision rests on and the repo text it clarifies or contrasts, give the resulting structure a formal model with falsifiable predictions, and specify the wave-3 program cell by cell. The weak side of the basket's convertibility undertaking is struck as a tranche ladder – successively deeper discounts sized to the reserve remaining – so it has no cliff for an attacker to time. Testability and modelling are the spine: nothing here ships on-chain that has not first won its catalogue cell.

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1 Purpose, goals, and how to read this document

1.1 The question

The Alberta Buck stabilizes BUCK against a defined basket of commodity tokens. Two kinds of flow act on that equilibrium:

- **Organic flow:** BUCK buyers and sellers arriving at varying rates and intensities for their own reasons – savers accumulating and spending, debtors converting mortgages and retiring, depositors entering and leaving the basket, and their waves and runs. Organic behaviour is to be SUPPORTED (deep liquidity at par, cheap execution, no gates) and REWARDED (the basket’s yield against demurrage).
- **Inorganic flow:** agents attempting to disrupt or corner the BUCK/TOKEN markets for profit or malice. Inorganic behaviour is to be DETECTED and HARVESTED: the attacker’s funds transferred to the BuckBasket investors’ ROI.

The basket’s issuance capacity and its reserves from prior profits are to stand ready to amplify that harvest for all position holders – while never holding positions that work AGAINST the restoration of equilibrium (a wall of basket positions that K must push the credit issuers through, or inventory that stalls the recovery it was bought to hasten). Ideally the cost of the basket’s positions is borne by the malicious actor, who must eat through them to recover their funds after disrupting the equilibrium.

1.2 The answer, in one paragraph

The system already holds equilibrium against organic flow and, when a private defender cast exists, against inorganic flow. But the harvest goal is not met, and until 2026-09-01 was unmeasured: at x4 an undefended attacker breaks the peg at zero cost to itself while depositors realize -\$2.0M to -\$2.7M; with the full cast, depositors are spared, the harvest (+\$1.8M realized) goes to private defenders, and the attacker’s -\$10.6M lands on third-party pool LPs and arbs. Harvest accrues to whoever is the COUNTERPARTY at the venue where the attack lands, and the basket is not that counterparty today. The remedy is a ladder target zone: private liquidity harvests day-to-day traffic near par; citizens’ latent credit – unleveraged insured assets behind BuckCredit limits, paying no demurrage while poised – issues into premiums and retires into discounts through an automated facility; and the basket stands behind both as a currency board in bundle units, with an announced UNLIMITED strong-side undertaking (sell BUCK for the bundle at par + eps) and an NAV-bounded weak-side undertaking (buy BUCK for the bundle at par - eps). K sees the SHADOW deviation – bvib plus the inventory the desk and the facility have absorbed or issued – so the slow lever mid-ranges every fast actuator it can see and the wall never blinds the controller. Restoration dominates harvest; the harvest is a small, certain margin on both edges plus a stress fee on duress exits, shared pro rata by a single receipt class; and the real return to investors is the excursion that no longer happens. A bvib-settled perpetual with a funding rate keyed to the shadow deviation charges the DURATION of positions held against the band – the missing long-side toll – and its telemetry sees the accumulation phase. Every element is specified as a catalogue cell with a pass/fail gate before any contract work.

1.3 Reading guide

- Section 2 maps the system as it stands, drawing on every design document, so the rest of the document can refer to mechanisms by name.

- Section 3 states the external critique, where it lands, and the basket-side measurement that reframes the program.
- Section 4 gives the theory: eight results from monetary economics, market microstructure and control, each tied to the repo text it clarifies or contrasts.
- Section 5 records the six decisions of record.
- Section 6 draws the resulting structure and its contract surface.
- Section 7 is the model: symbols, laws, bounds, and predictions the catalogue can falsify.
- Section 8 is the wave-3 program, cell by cell, with gates.
- Sections 9-12: risks and open questions, recommendations, glossary, references.

1.4 What this document settles against the other documents

document	what it settles	how this document uses or contrasts it
alberta-buck-ethereum.org	the contracts: BuckCredit, Buck, K, basket, demurrage	the sign convention, the funding factor, the
alberta-buck-ethereum-direct.org	the USD-free embodiment; bvib as process variable	the common-mode axis IS par; why the
alberta-buck-ethereum-basket.org	the heavy anchor; Shannon's demon; arbs work for free	the differential harvest vs the common-m
alberta-buck-rebalance.org / BASKET-REBALANCE.md	the directors: pairs/vrate, leash, persistence, timing	common mode from the same filter bank
BASKET-REDESIGN.md	custody, sell-high redemption, treasury split, regime table	the deflation branch; value conservation;
alberta-buck-operations.org / BuckBasketOps.sol	the desk: quadrants, three bounds, three runaways, the law	D1, D4; the desk is the executor of the u
BuckBasketFence.sol	K-scaled basket issuance; limitation list	D3 splits its concern (persistent stock) fr
alberta-buck-demurrage.org	2%/yr demurrage, Carrying accounts, the Jubilee	D6: poised capital is latent credit, not B
CREDIT-IMPROVEMENT.org	live credit limit, transfer lock, D8 unvetted insurers	latent credit as the wall's ammunition; L
alberta_buck/sim/EQUILIBRIUM.md	the campaign; approaches forward	approach (C) "direct stabilization chann
REBALANCING-EQ.org	the matrix, the catalogue, the roadmap	the instrument; recommendations 2, 4, 5
alberta-buck-ethereum-routing.org	BUCK-unaware arbs route through the basket pools	the benign workforce the treasury seeder
alberta-buck-platform.org	JS port decisions; the sim as specification	only demonstrated agents port; the cata

2 The system as it stands

This section is a map, not a re-derivation; every item names the document that settles it.

2.1 The common-mode loop: par, K, and the funding factor

- **Par is definitional.** In the direct embodiment BUCK measures itself against its basket: $\text{basketValueInBuck (bvib)} = \sum_i \text{basketAmount}_i \times \text{price}_i$ in-BUCK, read as a 600 s TWAP over the basket's own TOKEN/BUCK pools with spot fallback on a cold pool. Setpoint 1.0; error = 1 - bvib (alberta-buck-ethereum-direct.org; master doc, "Direct Variant").
- **Sign convention (decision of record).** bvib > 1: the basket costs more BUCK than it should, BUCK is CHEAP (a DISCOUNT excursion), error < 0, K FALLS, credit contracts, supply shrinks, BUCK recovers. bvib < 1: BUCK is DEAR (a PREMIUM excursion), K RISES, credit expands. Canary tests guard the direction.
- **K is the LTV cap:** $\text{creditLimit} = \text{activatedValue} \times K$. K0 = 0.75, rails (0, 0.95) so a railed K is still solvent; integral-dominant, a sustained 10% deviation spends rail-to-rail authority over $\tau_I = 90$ days with a 2% proportional kick; conditional-integration anti-windup at the rails; bumpless retune and re-prime; dT 1800 s (EQUILIBRIUM.md; BuckKControllerBase.sol). "Deliberately slow: K is a structural lever, not a market maker" (deploy.py).

- **The funding factor** is the on-chain counter-cyclical toll on insured issuance: $ff = \max(0, 1 + 10(b - p)/b)$, 2x at a 10% discount, ZERO once BUCK is 10% dear – issuance is cheapest exactly in a premium. It scales the pre-mint reserve on the insurance principal, not the gross mint; zero-premium credit is exempt. The sim's issue-rate arm ($ff_amp \times rate^{ff_pow}$) has no on-chain counterpart.
- **The theta law** (sim-only, the debtors' decision rule): deploy credit while the all-in execution discount $\leq \theta \times apr$; under an oversupply flood the marginal issuer stops at $\theta \times apr$, so the standing discount is set by the most aggressive unconverted debtor (~8% median, ~19.5% tail) – the flood is untested (REBALANCING-EQ.org recommendation 4).
- **Latent credit** costs nothing until drawn: activation happens only inside `Buck.mint`, premium is charged on present insured value at activation (CREDIT-IMPROVEMENT.org D6), demurrage accrues only on HELD BUCK, and a drawn position "becomes cheaper to close the longer it is carried" through Jubilee relief. D8 – unvetted insurers can write themselves zero-premium credit and mint for free – is the unclosed mainnet blocker.

2.2 The basket: custody, issuance, redemption, treasury

- The BuckBasket registers one TOKEN/BUCK Uniswap V3 pool per constituent and custodies a SINGLE FULL-RANGE Buck-owned LP position in each; it is the sole LP in every basket pool (`deploy.py pool_meta`). Full-range means the CPMM identity $NAV = 2 \times buckReserve$ and a 50/50 value split at any price (BASKET-REDESIGN.md 5.1).
- **DirectMint**: a depositor's TOKEN is paired with basket-minted BUCK at spot – 100% LTV, K-immune (`Buck.mintFromBasket` bypasses credit and funding factor; master doc #10) – and LP'd full-range; the depositor receives an ERC-721 receipt.
- **Redemption is sell-high**: the value claim $\theta \times NAV$ is drawn from the overweight pools by a closed form over the BUCK reserves, valued against a price-scaled fixed-quantity target; value conservation keeps the remaining holders' coverage ratio unchanged. The depositor is paid in TOKEN ONLY; BUCK profit ($Bw - R$) accrues to the treasury; a BUCK shortfall ($Bw < R$, the deflation branch) is covered from the depositor's TOKEN within `maxConversionLossBp` (1%), and $NAV < principal$ reverts (BASKET-REDESIGN.md 5, 6).
- The regime table (BASKET-REDESIGN.md 6) predicts treasury revenue "largest exactly under inflation, when the peg most needs defending" and a costly exit under deflation "so holders keep P outstanding"; the measured version (`plot_basket_split.py`) is that a single agent pushing BUCK 19% dear made every redemption take the deflation branch and cost depositors ~\$1.54M "rather than the agent causing it".
- The directors (`alberta-buck-rebalance.org`): `pairs` (deployed) runs a seven-scale tick-EMA ladder per leg, pairwise quorum 4/7 on the confirmed turn, leash 30/25%, matched-pair trades; `vrate` is the single-window engine; `factor` (trend regime) is Python-only. The COMMON MODE is the mean of the same ladders and was being discarded.
- The ops desk (`alberta-buck-operations.org`; `BuckBasketOps.sol`): the common mode drives four quadrants – Q1 absorb / Q3 supply as self-reversing range orders (repo), Q2 retire / Q4 issue outright under a 30-day-past-the-leash persistence test. Three bounds, each found by a runaway: per-leg 0.4% of the pool's BUCK reserve, position 10% of NAV with inventory-based escalation, cumulative 10% of NAV. The desk is capitalized by `capitalizeMonetary`

("a currency board can only defend with reserves it built up earlier") and may not fund itself from depositor TOKEN. `monetaryDeviationOffset()` exposes what the desk's inventory has taken out of K's error, deliberately not acted on. Default `enabled: false`; it has never run inside a matrix or catalogue arm.

- The fence (`BuckBasketFence.sol`): caps basket issuance at $K \times \text{TOKEN}$ value at TWAP in a $\pm 15\%$ band – "a continuum of resting limit orders" – so that K falling forces withdraw-and-burn and K rising lets it deploy. Its motivating measurement: K at its floor 21% of a 730-day run with the K-immune basket tranche at 47-52% of supply. Limitation 1: the budget is denominated in the unit that moves.

2.3 The excursion program and its instrument

REBALANCING-EQ.org is the design of record. The matrix (ten arms, all PASS at seed 0xA1BC), the raid arms, the portcast cast (settled physics, keyed RNG, no FatCredit; the JS-port cast), and the excursion catalogue: seven injections (none, dump, squeeze, grind-down, grind-up, spike, step) at day 365 of a 2-year window, against defender mixes (usdc = Q1 absorb, buck = Q3 supply, credit = Q4 issue / Q2 retire, basket = two-sided rotation in the bvib-defining pools, crb = the differential rebalancer, all), at defender scales $x1/x2/x4$ and injection budgets $x1/x2/x4$, measured by one response metric (eqmetrics: peak, AUC, recovery, K move and rail days, quadrant flows, P&L). Findings 1-11 stand. The catalogue is the falsifiable instrument this document extends; every cell is a `python -m alberta_buck.sim --experiment catalogue-<arm>.toml --backend pyrevm` run over the real contracts.

3 The critique, where it lands, and the basket's own side of the catalogue

3.1 The critique

Panoptic's "Yin (Carry) & Yang (Convexity) Require Each Other to Survive & Thrive" (X, 2026-08-27; saved copy) is the onchain transplant of the Artemis/Cole framing. In our terms:

A passive AMM LP is a carry trader – a seller of perpetual optionality who collects a fixed fee (the implied volatility) while eating adverse selection (the realized volatility). Onchain, the structural oversupply of LP positions suppresses IV below RV, and crypto is the most momentum-prone, fat-tailed asset class – so the profitable side is BUYING convexity from LPs, not being one. Carry books show negative skew ("sawtooth": steady fees, rare wipeouts); convexity books show positive skew ("stairstep"). A market is complete and stable only when both sides exist and both are PRICED.

3.2 The exposure and the measured evidence

The basket is the sole full-range LP in every constituent pool; its depositor-facing balance sheet IS that LP book; the controller's process variable is read from the same pools; every arb, whale slice and excursion trade in a basket pool has the basket (depositor L + treasury L) as counterparty. Static fee tiers (TOKEN/BUCK 30 bp, BUCK/USDC 5 bp): one implied vol for every regime and counterparty. The depositor is the residual claimant on both commodity moves and par deviation; the treasury holds a one-sided call.

Evidence already in the repo, not previously named: the \$1.54M deflation-branch expropriation (`plot_basket_split.py`); finding 6 (the MA-reversion differential rebalancer loses $\sim 6\%$ /yr trading

macro drift as reversion – $RV > IV$ on the differential axis); finding 11 (foresight turns the same book to +\$4-5M – the informed convexity buyer's profit is real and the quoting side pays it); the ops desk losing on **drift** (2/9 seeds) and the law "a lagged operator is a profitable destabilizer"; the fence test's "genuine impermanent loss that swamps the fee income"; and the absence of any LVR/markout accounting (`snapshot.py` LP APR is gross fee yield).

3.3 The basket's own side of the catalogue (measured 2026-09-01)

No catalogue summary carries a BuckBasket, treasury or depositor column; "attacks punish themselves superlinearly" (finding 8) is a statement about PRIVATE defenders. The frames carry **basketNav**, **treasuryBuck** and **dmProfitUsd**, so the basket side was computed from the raw vectors (units par-USD; end = day 730; deltas only where the mix's pre-history is byte-identical to the undefended cell):

arm	mix	bvib in window	NAV USD end	treasury end	depositor realized end	defender real / marked	whale	K e
dump x1	none	0.99-1.13	47.2M	230k	+0.81M	–	-0.01M	0
dump x1	usdc	0.99-1.09	46.4M (-0.75M)	220k (-9k)	+0.90M (+0.09M)	+0.26 / +0.26	-0.01M	0
dump x1	basket	0.99-1.11	47.6M (+0.43M)	222k (-8k)	+0.97M (+0.16M)	+0.01 / -0.60	-0.01M	0
dump x4	none	0.63-1.12	45.3M	255k	-1.99M	–	0.00M	0.9
dump x4	all	0.95-1.32	50.5M	277k	+0.67M	+1.81 / +3.78	0.00M	0
squeeze x1	none	0.90-1.01	47.8M	223k	+0.94M	–	-0.87M	0
squeeze x1	credit	0.90-1.05	48.0M (+0.16M)	228k (+5k)	+0.89M (-0.05M)	0.00 / +0.63	-0.89M	0
squeeze x4	none	0.69-1.01	47.8M	257k	+0.23M	–	-9.13M	0
squeeze x4	all	0.76-1.04	51.9M	237k	+0.68M	+1.50 / +4.93	-10.59M	0
grind-dn x4	none	0.61-1.02	44.9M	424k	-2.72M	–	0.00M	0.9
grind-dn x4	all	0.97-1.02	49.1M	211k	+0.62M	0.00 / +1.74	0.00M	0

(* railed for the whole tail.)

Four readings, each load-bearing below:

1. **The x4 "dump" and "grind-down" failures are SQUEEZE failures.** The bvib minima (0.63, 0.61) come from the whale's \$30-40M accumulation and re-accumulation legs, which drive BUCK 35-40% dear and rail K at 0.95 – the controller begging for supply the K-scaled issuers cannot deliver. The premium flank is the weak flank, as wave 2 said and as corner theory predicts (4.5).
2. **Undefended malice is free at x4.** The whale ends at \$0.00M in both failed arms while the peg is broken for a year.
3. **Depositors are the residual loser, and the treasury's call pays MORE when the economy is broken** (424k vs ~210k) while depositors realize -\$2.7M.
4. **Defense is a transfer avoided, not a harvest captured.** At x1 the USDC defenders' +\$0.26M coincides with the basket NAV ending -\$0.75M versus the undefended cell: the pools sold the dump cheaply to the defenders. The whale's -\$10.6M at x4 squeeze lands mostly on the BUCK/USDC pool's LPs (SimLP, i.e. third parties) and the arbs, because that is the venue the whale trades in.

The system holds; the basket investor is at best protected and at worst expropriated; the harvest accrues to whoever is the COUNTERPARTY at the venue where the attack lands.

3.4 Where the critique does not land: manufactured reversion

Panoptic's $RV > IV$ inversion is an empirical claim about UNANCHORED, momentum-prone assets. The bvib axis is engineered to mean-revert: K plus the poised cast manufacture the reversion a TradFi option seller merely bets on. Attacker P&L is superlinearly negative on squeezes (-\$0.87M / -\$3.06M / -\$9.13M at x1/x2/x4), defenders are paid countercyclically, and the excursion agents' `min_edge` gate means they refuse to sell convexity below a 1% floor premium. So the two axes split:

axis	vol regime	correct posture
common mode (BUCK vs basket) differential (TOKEN i vs j)	manufactured reversion genuinely trending	SELL convexity – priced, never free; own the edges patience (the director), paid foresight, hedges

This is exactly the split `alberta-buck-ethereum-basket.org` makes between the anchor (the mass that resists being moved) and the harvest (Shannon's demon on reverting constituents): the basket SELLS reversion insurance on the axis it controls and BUYS reversion on the axis the physics controls, and the critique bites only where those two are confused – a per-pool full-range position sells both at once.

3.5 Corrections to v1.0

1. **Reference-price contamination.** Common-mode markout must be measured against PAR (reversion to 1 at horizon D); the truth-implied TOKEN/BUCK reference (via BUCK/USDC) is the pool a BUCK dump moves and serves the differential ledger only.
2. **$\sigma^2/8$ overstates LVR where reversion is manufactured;** realized markout on the common mode, analytic LVR only on the differential.
3. **Symmetric fee surges slow recovery:** only a DIRECTIONAL policy is admissible.
4. **The ops desk was under-credited:** v1.0's options C and F existed as `BuckBasketOps`; the gap is that it never ran in the loop.
5. **The venue was missing:** v1.0 never asked where attacks land.

4 Theory: eight results and the repo text each one clarifies or contrasts

Each result is stated, then tied to the document it explains, sharpens or overturns. These are the load-bearing arguments behind Section 5.

4.1 Carry and convexity: what the basket sells and what it buys

Kyle (1985) and Glosten-Milgrom (1985): a standing quote is a free option to informed traders and a paid service to uninformed ones; the quoter's profit is the uninformed flow's spread minus the informed flow's markout. Milionis, Moallemi, Roughgarden and Zhang (2022) put the AMM case in closed form: loss-versus-rebalancing $\sigma^2/8$ per unit time against a martingale price. Panoptic's contribution is the empirical inversion onchain ($RV > IV$) and the framing that carry and convexity books need each other to be PRICED.

Clarifies `alberta-buck-ethereum-basket.org`: "the world's arbitrageurs, working for free" is true of the BENIGN flow – the uninformed cycles that repin the pools and pay the fee – and false of the informed flow, which the same pools sell the basket's optionality to. The basket doc's yield thesis

(Shannon's demon, +322 to +496 bp/yr measured in `BASKET-REBALANCE.md`) is a DIFFERENTIAL harvest that requires reversion; the COMMON-MODE convexity the same full-range positions sell is a different exposure with a different owner of the reversion (K), and the two must be separated in the position structure. **Contrasts** the routing doc's sim, which measured that BUCK-unaware arbs genuinely route through the basket pools (~1100 routed trades vs ~120 whale snaps) – that flow is the workforce the treasury seeder of D5 exists to lubricate, and it must stay cheap.

4.2 Target zones: a credible band stabilizes inside itself

Krugman (1991): with a credible commitment to intervene at the edges of a band, the exchange rate inside the band is LESS volatile than the fundamentals – the honeymoon S-curve – because speculators front-run the edge. Bertola and Caballero (1992): with IMPERFECT credibility the effect reverses; expected realignment makes the rate more volatile near the edges. The asymmetry is empirical: Hong Kong's strong-side Convertibility Undertaking (unlimited HKD sales at 7.75, 2005) has never been abandoned and never been beaten; the SNB's EUR/CHF floor (2011-2015), mechanically identical, was abandoned because unlimited issuance accumulated a balance sheet approaching GDP in a foreign asset against a FUNDAMENTAL pressure it could not make revert.

Clarifies `BuckBasketFence.sol`: "a continuum of resting limit orders" is a band without announced edges and with a budget "denominated in the unit that moves" (limitation 1); D3 gives the band announced, bundle-denominated edges and keeps the fence for what it is good at – K-scaling the PERSISTENT depositor stock. **Contrasts** the ops doc's reserve doctrine ("any bound it publishes will be probed – the 1992 ERM crisis is the standing lesson"): that is the WEAK side's truth, where the defender spends finite reserves; on the STRONG side the defender issues its own liability, and the correct doctrine is the opposite – announce it unlimited. **Explains** the catalogue: the x4 failures are all on the premium side because the weak side of the current design (private absorbers, the desk's TOKEN book) is reserve- bounded AND the strong side (issuance) is K-scaled and slow.

4.3 The backstop at a penalty rate

Bagehot (1873): in a panic, lend freely, at a penalty rate, on good collateral. The penalty rate is what keeps the backstop from being used in normal times and pays it when it is.

Clarifies the design owner's ruling that "the BuckBasket isn't a market-making liquidity provider, but a stabilizing backstop": the basket's depth sits OUTSIDE the private band, at a small fixed premium eps, and earns that premium for depth-on-demand. **Contrasts** the ops doc's measured trade-off ("stabilization and extraction trade off": 10-day signal peak 1004 bp / desk +\$623k; 80-day 1344 bp / +\$746k; outright inverts past 80 days): the penalty rate is a FIXED small premium, never a stress-scaled extraction, and the recovery is never held hostage (D1).

4.4 Mid-ranging: two actuators, one process variable

Shinskey's valve-position (mid-ranging) control: when a fast small actuator and a slow large one act on the same variable, the fast one controls the process variable and the slow controller does NOT look at the process variable – it looks at the fast actuator's POSITION and drives it back to mid-range. Two controllers on one variable without this separation oscillate or wind up. The Federal Reserve's pre-2008 open-market desk is the monetary instance: the FOMC sets the target; the desk holds the funds rate there with daily temporary repos; ever-larger temporary operations in one direction are the signal for an outright purchase. The FX literature (Jurgensen 1983; Dominguez-Frankel 1993) is the negative result: sterilized intervention – the desk acting while the policy lever is unaware – is weak and transient.

Clarifies `alberta-buck-operations.org` runaway one ("the operator suppresses its own alarm") and `BuckBasketOps.monetaryDeviationOffset` ("exposed rather than acted on, because the right response is a controller-side choice"): the controller-side choice is mid-ranging (D4). **Clarifies** `deploy.py`'s "K is a structural lever, not a market maker; it glides over months while private demand does the fast stabilization" – that is a frequency separation with the coupling left unspecified; D4 specifies it. **Overturns** v1.0's "K-gradient gate" question: the desk and the facility do not gate on dK; their books ARE what K mid-ranges.

4.5 Corners and the ability to deliver

A corner breaks when the cornered asset can be delivered in any size at a known price. The Hunt silver corner (1980) was broken by changing the rules mid-game – effective, and corrosive to trust in the venue. Hong Kong's strong side breaks any HKD corner by standing rule. The SNB shows the one way unlimited delivery fails: when what it accumulates is a liability you cannot make revert.

Clarifies the ops doc's "what only the contract can do – Q4 is not capped by K" and the fence's "the most K-immune half of the float": the K-immune `mintFromBasket` is the system's ONLY anti-corner power, and it must not be blunted by K-scaling; D3 resolves the conflict by duration (persistent stock K-scaled; standing facility K-immune and self-reversing). **Explains** why the basket's unlimited undertaking is SAFER than the SNB's: it sells BUCK dear for TOKEN that is cheap in BUCK, on an axis where par is definitional and K forces the reversion, and Q2 closes the round trip. The only path to the SNB's fate is governance abandoning par.

4.6 The elastic currency and the real-bills doctrine

The Federal Reserve Act (1913) created an elastic currency: money supply expands when banks discount real bills at the window and contracts as the bills self-liquidate. Its historical flaw is procyclicality (credit expands as collateral values rise), which a counter-cyclical lever exists to correct.

Clarifies the design owner's D6 ruling: the poised capital is NOT BUCK but LATENT CREDIT – unleveraged insured assets behind `BuckCredit` limits – issued into premiums and retired into discounts. That is a privately discounted elastic currency with K as the haircut and the funding factor as the discount rate, and the on-chain funding factor already makes issuance cheapest in a premium. **Contrasts** `alberta-buck-demurrage.org` (demurrage punishes idle BUCK) with the demand for standing defense capital: no conflict, because the standing capital is not BUCK. **Sharpens** `EQUILIBRIUM.md` approach forward (C), "a direct TOKEN/BUCK stabilization channel for savers" – the auto-pay-down facility of D6 is that channel, on the credit side.

4.7 Swing pricing: the exiting investor pays for the liquidity they consume

The Reserve Primary Fund broke the buck in September 2008 and those who STAYED bore the loss; the SEC's 2014 and 2023 reforms made liquidity fees on stressed redemptions mandatory for institutional prime funds. Liquity's Stability Pool (opt-in, paid) versus passive LPs in the UST/3CRV pool of May 2022 (conscripted, expropriated) is the same lesson in DeFi.

Clarifies `BASKET-REDESIGN.md`'s regime table – its deflation branch is the pre-2014 money-market fund, and the measured expropriation (\$1.54M; -\$2.7M at x4) is the proof – and the ops doc's "the desk may not fund from depositor TOKEN" (it refuses to conscript explicitly while full-range custody conscripts implicitly). D2's stress fee is the fix the one industry that learned it the hard way adopted.

4.8 Reserve adequacy and the two generations of crisis model

Krugman (1979) / Flood-Garber (1984): a peg defended with finite reserves against a persistent fundamental misalignment fails at a computable date. Obstfeld (1986, 1994): with costly defense, attacks can be self-fulfilling. Both say the same thing about the weak side of a band: its credibility is its reserve relative to the plausible outflow over the horizon on which the slow lever can correct.

Clarifies the ops doc's "reserves are finite and the bound is public" and the catalogue's x4 grind-down with the full cast (parity held, rail margin spent: K 0.936, 17% railed) – that cell is the empirical adequacy frontier. **Sets** the sizing rule of 7.5.

The first-generation model has a constructive corollary: the attack is timed to the DISCONTINUITY – the moment a flat edge backed by a finite reserve stops being honored. Remove the discontinuity and the timing argument has nothing to bite on. A defender whose bid price declines smoothly with its remaining reserve is never "out"; it is only ever "dearer", and the attacker's marginal cost rises with every unit sold into it. This is the shape of every constant-product bid (the catalogue's superlinear squeeze cost is this property of the BUCK/USDC pool), of Curve's stableswap between its flat centre and its convex tails (the amplification parameter A is the dial), and of a central bank that sells reserves by declining-price auction rather than at a fixed rate. **Clarifies** the weak-side design: the design owner's tranche ladder (D3 refinement) is the discrete form, and the flat edge of a classical target zone is its delta $\rightarrow 0$ limit – the one limit with a cliff.

5 Design decisions of record (2026-09-01/02)

Six questions were open after the audit. Each was worked through one historical and one synthetic example with the design owner; the answers are decisions of record.

5.1 D1. Restoration dominates; the backstop earns a fixed premium

Question. When harvest and restoration conflict, which dominates?

Historical. Hong Kong, August 1998: the "double play" (short HKD to force rates up under the currency board; short Hang Seng futures to profit from the crash). The HKMA spent HK\$118bn over ten trading days buying 33 constituents – a bid wall in the venue where the attackers' PROFIT was booked – lifting the index from $\sim 6,700$ to $\sim 7,800$ into the August expiry; the shorts were forced to cover into the wall (harvest at ENTRY). The unwind was restoration-first: held ~ 15 months, then disposed through the Tracker Fund (Nov 1999; HK\$33.3bn first tranche to 180,000 retail investors; $> \text{HK\$}100\text{bn}$ recouped over three years; $\sim \text{HK\$}100\text{bn}$ profit) – from the RECOVERY, into organic demand, never sold back to the attackers at a premium. Then the commitment was hardened (the September 1998 technical measures); no comparable attack since. Sterling 1992 is the contrast: reserves spent against a misaligned fundamental. The difference is whether the defender can MAKE the reversion; par is definitional and K manufactures it.

Synthetic. The ops doc's measurement-scale ladder (above): extraction bought $+\$123\text{k}$ of desk profit for $+340$ bp of peak. A $\sim 20\%$ deviation held a year swung depositors' realized P&L by $\sim \$2.7\text{M}$: the organic cost of prolonging an excursion is of the order of the ENTIRE harvest.

Decision. Restoration dominates. The conflict exists on ONE of four legs:

side	entry leg	exit leg
squeeze (BUCK dear)	Q4 sell issuance at a premium: harvest AND restoration	attacker dumps back $\rightarrow$ Q1/Q2 buy cheap, burn: harvest
dump (BUCK cheap)	Q1 buy the dump at a discount: harvest AND restoration	selling inventory back at a premium: harvest AGAINST r

Harvest is taken on the three coinciding legs. The fourth is unwound TraHK-style – rate-limited into organic flow at par over K’s correction horizon – or burned (Q2) when persistence says the move was real. Extraction is not an objective function.

Refinement. The basket is a stabilizing backstop, not a market maker, and earns a SMALL FIXED premium eps for depth-on-demand (Bagehot). Private LPs inside $\pm x\%$ harvest the traffic; the basket’s deep liquidity sits outside. A recovering dump reaches the basket’s ask side only if it OVERSHOOTS into a premium, so inventory bought on the bid side is sold at a premium only when the market pays it; the remainder is disposed at par. Spot tails (V3 now, V4 directional-fee tails later) are the facility; a perpetual is not (a trader needing depth wants conversion, not a derivative – D5).

5.2 D2. Single receipt class; a stress fee on duress exits

Question. May depositor liquidity be wall ammunition, and who bears the cost among depositors? (Depositors already ARE the wall – involuntarily, unpaid, at every price.)

Historical. Liquity’s Stability Pool (opt-in, paid, held through 2022); the UST/3CRV LPs (conscripted, expropriated); money-market fund reform (the exiting investor pays; a fee is not a gate).

Synthetic. Defended x4 cells show the return to depositors of a paid wall: +\$0.67M instead of -\$1.99M.

Decision. A SINGLE receipt class – every depositor pro rata in band and tails; the harvest (undertaking margins, stress fees, rebalancing premium) shared pro rata. A STRESS FEE on duress exits: a redemption landing in the deflation branch during a measured excursion pays a fee proportional to the deviation, credited to remaining holders. "If you need to exit under duress, you take the hit." Held long enough, K drives bvib back to par and rebalancing garners the reversion premium. Compatible with "always redeemable, no gate".

Caveat of record. "IL does not bite the long-term holder" is true by construction on the common mode and true on the differential mode only to the extent constituents revert – the catalogue’s permanent NRG step leaves a residual. The promise rests on the admission discipline of alberta-buck-ethereum-basket.org ("only civilizational basics"), not on the LP mechanics.

5.3 D3. Two issuance classes; the Convertibility Undertakings

Question. The x4 failures are corners the K-scaled issuers cannot answer; the only actor that can deliver BUCK on demand is the basket via K-immune `mintFromBasket`; the fence wants basket issuance K-scaled.

Historical. Hong Kong’s strong side (unlimited, standing, never beaten); the SNB floor (unlimited, abandoned – the balance sheet it accumulated); the Hunt corner (rules changed mid-game). Standing rules beat emergency ones.

Synthetic. Squeeze x4 undefended: whale -\$9.13M, bvib 0.69, K railed all year. In the isolation runs Q4 fired 2 and 10 times and Q4+Q2 was what the external agent could not do. The fence’s coupling has the SAME sign as the ops quadrants; its concerns – a persistent K-immune STOCK and a budget in the unit that moves – do not apply to self-reversing issuance retired by Q2.

Decision. Two issuance classes, distinguished by DURATION:

1. **Depositor issuance** (persistent stock: DirectMint partner BUCK living as long as the deposit) – K-scaled via the fence once fence-in-the- loop passes its cell, so K governs the stock the economy carries.

2. The Convertibility Undertakings (standing, announced, UNLIMITED in aggregate on the strong side):

- **Strong side.** Anyone may buy BUCK from the basket at $\text{par} + \text{eps}$ paying the basket BUNDLE in `basketAmount` proportions; or at $\text{par} + p \cdot \text{eps}$ paying ANY token with a conversion path known to the basket at deposit time (USDC, BUCK, WBTC, PAXG, ...), the basket spot-converting the incoming asset atomically into its underweight constituents by its rebalancing metrics. p is a convenience factor ($\sim 2 \times$ the route conversion fees; may scale with size for impact). "Unlimited" is the bundle path; the convenience path is bounded by conversion capacity (`maxLegBp` on the route, so the redemption guard never bricks).
- **Weak side.** Anyone may sell BUCK to the basket for the bundle at $\text{par} - \text{eps}$, or for a single token at $\text{par} - p \cdot \text{eps}$, the mix chosen by the basket's under-/over-weight metrics (sell-high) and converted to one TOKEN atomically in the redemption transaction. Reserve-bounded by NAV; the stress fee of D2 composes under duress.
- **The weak side is a LADDER, not an edge** (refinement, 2026-09-02). Because the weak side's reserve is finite and "must outlast K's correction", its bid price scales DOWN with the reserve capacity remaining: a set of positions sized and priced so that liquidity is always available, but successive tranches are available at successively poorer rates. Tranche k is struck at a discount $\text{eps} + (k-1) \cdot \text{delta}$ and sized at a fraction ϕ of the reserve remaining when it is struck, so the reserve depletes geometrically and never reaches zero at a finite price. There is no cliff for an attacker to time (4.8), the attacker's marginal cost rises with cumulative size by construction, and the deviation itself becomes an honest gauge of reserve stress – which is the coupling to K the shadow term wants (7.4). $\text{delta} \rightarrow 0$ is the flat edge; a fixed delta/ϕ ratio is the constant-product bid (the shape the basket's full-range position already has); Curve's stableswap amplification is the interpolation. The same rule applies to the STRONG side's capacity-bounded convenience path (the premium $p \cdot \text{eps}$ rising as the epoch's conversion capacity is consumed); only the unlimited bundle path is flat.
- K-immune by design (fires only at a premium/discount, exactly when the funding factor already says issuance/retirement is cheap); accounted to the monetary book; retired/unwound by Q2/Q1 under the persistence test; bundle-priced so it carries no differential exposure.
- **Announced unlimited.** A published cumulative cap would be probed; what unlimited issuance accumulates here is reversion inventory on a controlled axis, and Q2 guarantees the balance sheet shrinks back.

Consequence. A currency board in bundle units: while both sides are honored, `bvib` is pinned inside $[1/(1+\text{eps}), 1/(1-\text{eps})]$ and Krugman's honeymoon applies inside. The $x4$ squeeze re-run: the pool price cannot exceed $\text{par} + p \cdot \text{eps}$; the whale's \$40M buys BUCK from the basket; the basket books $\sim 2 p \cdot \text{eps}$ over the round trip; the excursion never exceeds the band; K barely moves; depositors never enter the deflation branch. Deterrence replaces harvest – and for basket investors (from $-\$2.7\text{M}$ to $\sim +\$1\text{M}$) deterrence wins.

5.4 D4. K sees the shadow `bvib` (mid-ranging)

Question. The desk absorbs a dump, `bvib` improves, K sees less error and stops tightening, the desk is stranded; under D3 the desk's book is unbounded on the strong side. Sterilized or unsterilized?

Historical. The FX-intervention literature; the Fed desk's temporary- to-outright rule; mid-ranging control (4.4).

Synthetic. Runaway one (49.9% of pool depth, profitable in 7/9 seeds, never escalated); the 90-day run (1.25M BUCK held while K still fell 0.745 -> 0.584 – "not suppressed AT THAT SIZE").

Decision. MID-RANGING. K's process variable becomes the shadow deviation

$$bvib_shadow = bvib + \lambda * (\text{net monetary inventory} / \text{reference depth})$$

with BUCK ABSORBED under the weak side counting positive and BUCK ISSUED under the strong side counting negative – bvib as it would have been without the desk. Equivalently, K is tasked with driving the desk's inventory to zero. Three things follow: the desk's position is never a bet on "K might keep forcing"; Q2 burns need no special treatment (burned inventory leaves the book and the shadow term drops to zero – the temporary/outright and sterilized/unsterilized distinctions are the same distinction); and it IS the achieved-action anti-windup the ops doc asked for. The design owner's statement of intent: "even if we manage to raise bvib back to parity, K must reflect the error we've damped by our action."

5.5 D5. No USD exposure; the treasury seeds BUCK/USDC temporarily; the perp is a funding-rate lever

Question. The attacker's losses land on the BUCK/USDC pool's LPs. Must the basket own that venue, taking BUCK/USD exposure it has no policy control over?

Historical. Counterparty vs venue (COMEX earned fees from the Hunts and harvested nothing; the HKMA stood as counterparty); Hong Kong after 2005 (attacks stopped: deterrence replaces harvest); the cheap band-break bet and the currency board's automatic interest-rate mechanism (overnight HIBOR ~280% in October 1997; the aggregate balance from ~HK\$337bn to ~HK\$45bn in 2022-23 until the shorts gave up), and its 1997 flaw – the rate spike crushed every organic borrower too.

Synthetic. Under D3 the basket owns the EDGES; it need not own the pool. What the undertakings do not do: cap the DURATION of a position against par (inside the band a corner is cheap to hold – a short-BUCK bettor draws credit and pays premium, funding toll and K's contraction; a long-BUCK bettor pays only 2%/yr demurrage; the squeeze flank is weak because holding a squeeze is cheap), or see the ACCUMULATION phase.

Decision.

1. The basket takes NO USD exposure and does not own the BUCK/USDC venue; it becomes a private venue for inside-the-band traffic.
2. **The treasury seeds BUCK/USDC depth TEMPORARILY at launch**, to lubricate the BUCK-unaware arb route that repins the TOKEN/BUCK pools via BUCK/USDC: concentrated positions from the CURRENT price toward the IDEAL (the bundle's USD value at par, read from the truth pools) and not beyond – private LPs supply depth beyond ideal. A single-sided V3 position from current toward ideal is always funded with the asset the CORRECTIVE arb wants to take (BUCK if the price must rise, USDC if it must fall), so it is the counterparty to corrective flow by construction, self-reversing, and withdrawn as private depth arrives – `BuckPoolInvestorAgent`'s placement with a different objective (lubricate, not earn). Orientation is the "silent killer" of the ops doc; verify in sim.
3. **The perp is a bvib FUNDING-RATE LEVER, not a trading venue.** Settled on bvib (par-denominated), never on BUCK/USD; funding set by the system as a function of the shadow deviation; the monetary book as the house, mid-ranged by K. Hong Kong's automatic

interest-rate mechanism TARGETED: it charges positions held against the band and nobody else. Its OI, basis and funding are the accumulation-phase telemetry; its payoff is bounded by the band. Built as a sim agent first, or an installed contract if easily obtained (wave 3f).

5.6 D6. Demurrage stays; poised capital is latent credit; the auto-pay-down facility

Question. Should capital verifiably parked in defense posture be rebated its demurrage?

Historical. Woergl (demurrage's target is the idle hoard); interest on reserves vs negative deposit rates; catastrophe bonds (coupon for committed capital, funded by premiums); the elastic currency (4.6).

Synthetic. Which poised capital EARNED in the catalogue: the USDC absorbers (no demurrage) realized +\$0.26 / +0.28 / +0.39M; every BUCK-holding mix bled; the credit base was scale-stable, +\$0.63M marked, and paid nothing while poised because it holds no BUCK.

Decision (design owner). Demurrage on poised BUCK is not something that should happen – because the poised capital is NOT BUCK. It is LATENT CREDIT: a hopefully revenue-generating insured asset (farms, homes, businesses – the plentiful pent-up un-leveraged asset base) behind a BuckCredit contract with an unused limit, ready for deployment, with no BUCK sitting in an account accruing demurrage. When BUCK is at a premium, holders issue against their limit and buy assets with the temporarily too-valuable BUCK; at a discount they buy BUCK cheap and pay their credit down. Four rules:

1. **No readiness rebate.** Demurrage stays; BUCK parked in the basket earns the single-class harvest (the basket doc's "yield against demurrage" made concrete); weak-side defense capital is TOKEN/USDC/NAV and never paid demurrage; the strong side is issuance, not inventory; the monetary book's own BUCK pays demurrage as an internal transfer to the Jubilee.
2. **eps is LARGER than a backstop alone would choose**, leaving an inner zone where ordinary citizens with available credit "get in on the deflation action" and "buy back their credit limit at a temporary inflation moment". The private latent-credit cast is the middle of the ladder; the basket is the wall behind it.
3. **The auto-pay-down facility.** Automated, user-parameterized machinery attached to a BuckCredit position: issue against the limit when $bvib < 1 - x$ (premium) and place the proceeds (default: a basket deposit under the single receipt class); when $bvib > 1 + y$ (discount), redeem/buy BUCK cheap and repay (retirement happens on the BUY; the burn closes the line). Parameters: x , y , maximum fraction of limit to deploy, target asset. **ExcursionCreditArbAgent's** Q4/Q2 logic as an on-chain facility, pre-priced by the funding factor – an offset mortgage that draws when your currency is dear and repays when it is cheap.
4. **Facility utilization enters K's shadow term** (7.3): deployment in a premium is net issued BUCK, retirement in a discount is net absorbed, exactly like the desk's book. Weak-side extension beyond NAV, if ever wanted, takes the cat-bond form (capped vault, only-toward-par, coupon from realized eps and fee-surge income), never a rebate.

6 The resulting structure

6.1 The ladder target zone

$bvib$ (basket in BUCK) who acts, and how

> 1/(1-eps) sloped, no cliff WEAK-SIDE LADDER: the basket buys BUCK for the bundle at par - eps, then at successively deeper discounts as its reserve depletes (tranche phi of the remaining reserve per delta step; per-leg maxLegBp) 1+y .. 1/(1-eps) discount zone latent credit RETIRES (the facility); private Q1 absorbers; private LPs' tails; the desk's Q1 range orders 1-x .. 1+y the band private LPs harvest traffic; the treasury seeder (BUCK/USDC, launch only); the honeymoon effect 1/(1+eps) .. 1-x premium zone latent credit ISSUES (the facility); private Q3 suppliers; the desk's Q3 < 1/(1+eps) cannot persist STRONG-SIDE UNDERTAKING: the basket sells BUCK for the bundle at par + eps (UNLIMITED; convenience path par + p*eps, bounded by conversion capacity)

across all zones: K mid-ranges the desk + facility books (bvib_shadow); the bvib perp charges the DURATION of positions held against the band; the directional dynamic fee prices raid-speed displacement at tick speed; the stress fee prices duress exits.

Deterrence is the product; harvest is the by-product; both accrue to the single receipt class pro rata.

6.2 Where each flow lands, and who is the counterparty

flow lands in counterparty harvest

organic small (savers, the band: private LPs, private LPs fees (theirs) debtors, DM churn) BUCK/USDC, TOKEN/BUCK organic large (a refinance, the band then the zone private + facility par+/-x, cheap a depositor wave) then, if needed, the edge then the basket eps (basket) inorganic squeeze strong-side undertaking THE BASKET p*eps entry + (once past par + p*eps) p*eps exit inorganic dump weak-side ladder THE BASKET (NAV) p*eps+ entry, (once past par - p*eps) + private Q1 organic unwind inorganic accumulation the perp (OI/funding) and the house (monetary funding (quiet phase) the band book) x duration raid-speed block the directional fee at the LP position surge fee tick speed (single class) duress exit the deflation branch remaining holders stress fee

6.3 The contract and agent surface (new and changed)

element	where	change
strong-side undertaking	BuckBasketOps (or a sibling shell)	buyBuck(bundle) at par+eps; buyBuck(token, amount)
weak-side ladder	same	sellBuck(bundle) at the current tranche's discount
shadow view	BuckBasketOps	shadowValueInBuck() = bvib + lambda_d * offset
controller input	BuckKControllerDirect	a Shadow variant whose _readReferences reads shadowValueInBuck()
persistence-driven retirement	PairsRebalanceDirector / BuckBasketOps	existing Q2/Q1 escalation applied to the undertaking
stress fee	BuckBasketProRata._redeem	deflation-branch fee ~ deviation, credited to remaining
the facility	new contract, user-parameterized	draws against a BuckCredit line into a basket deposit
treasury seeder	sim agent, then treasury program	BuckPoolInvestorAgent variant: current -> ideal, launch
the perp / house	sim agent + Python venue ledger, then contract	funding = g(bvib_shadow); OI cap; house = monetize
directional dynamic fee	sim pool adapter, then a v4 hook behind the venue seam	base fee on the corrective side; RV- and size-scaled

6.4 What this supersedes in earlier documents

- REBALANCING-EQ.org recommendation 2 (buy back rail margin for the growth arms): the strong-side undertaking is the supply valve the admission frontier lacked; the "top-up borrower" remains valuable but is no longer the only lever. Recommendation 4 (the oversupply flood) is now ALSO the D8-facility question (9.1). Recommendation 5 (fence-in-the-loop) narrows to the depositor stock. Roadmap item 7 ("fold the winning knobs into the BuckBasketOps/Fence revision") is realized as D3-D4-D6.

- **alberta-buck-operations.org**: "the desk may not fund from depositor TOKEN" is superseded by D2 (depositors are the wall, paid, with a fair exit); the three bounds remain and scale with the shadow deviation (9.4); **monetaryDeviationOffset** is promoted from a view to the controller's input.
- v1.0 of this document: options C and F replaced by D3 and D6; D2 (the perp) rewritten as the funding lever; the K-gradient gate cells replaced by the lambda sweep.

7 The model

7.1 Symbols and conventions

symbol	meaning
bvib	basketValueInBuck: BUCK cost of one bundle (TWAP 600 s over the basket pools)
e	controller error, $1 - \text{bvib_shadow}$ (positive: BUCK dear; negative: BUCK cheap)
K	the LTV multiplier; $\text{creditLimit} = \text{activatedValue} \times K$; rails $[0, 0.95]$
τ_I	the controller's integral time (~ 90 d); the correction horizon
eps	the undertakings' bundle premium/discount; the band half-width in bundle units
δ, ϕ	the weak-side ladder: discount step per tranche; tranche size as a fraction of remaining reserve
ρ	weak-side reserve fraction remaining, R / R_0
p	the convenience factor on single-token undertakings ($\sim 2 \times$ route fees)
x, y	the facility's issue (premium) and retire (discount) thresholds
D	reference depth for the shadow term (BUCK reserve of the basket pools)
I_d, I_c	net inventory of the desk and the facility: absorbed - issued (BUCK)
$\lambda_{\{d,c\}}$	shadow gains
u	facility utilization: drawn / limit over participants
R_w	weak-side reserve: NAV (TOKEN) deployable at par - eps
C_{leg}	per-leg conversion capacity: $\text{maxLegBp} \times \text{pool BUCK reserve per epoch}$
S	attack size (BUCK)
f	perp funding rate (per day), $f = g(e)$

7.2 The plant, reduced

The ops doc's `monetary_ops.py` abstraction is adequate here: one common-mode price with constant-product impact against depth and an OU reversion whose rate is what the private cast and K supply. A flow F (BUCK sold, positive) moves bvib by $\sim F/D$ at impact; the cast reverts it on its threshold-gated timescales; K moves supply through `creditLimit` on τ_I . Nothing below depends on the reduced form except the qualitative claims; the catalogue runs the real contracts.

7.3 The controller and the shadow term

$\text{bvib_shadow} = \text{bvib} + \lambda_d * I_d / D + \lambda_c * I_c / D$
 $e = 1 - \text{bvib_shadow}$
 $dK/dt = K_p * de/dt + K_i * e$ (conditional integration at the rails)

Signs: BUCK absorbed under the weak side (or retired by the facility) makes I positive, raises bvib_shadow, makes e more negative, and K falls harder – the contraction the absorption was betting on. BUCK issued under the strong side (or by the facility) makes I negative, lowers bvib_shadow, makes e more positive, and K rises – restoring the headroom the fast actuators spent. A Q2 burn removes the inventory from I_d ; the shadow term drops; K sees the real bvib; no double count. $\lambda \rightarrow 0$ is today's controller; λ large makes K chase every temporary operation. The mid-ranging claim: with λ at its knee, the desk's inventory returns to zero within $\sim \tau_I$ of any excursion that K can correct, and grows without bound only when the deviation is structural – which is the signal the persistence test uses to escalate to Q2/Q4.

7.4 The undertakings

Strong side: the basket sells 1 BUCK for $(1 + \text{eps})$ bundles' worth of TOKEN (bundle path) or $(1 + p^*\text{eps})$ bundles' worth of any convertible token, converting atomically. Whenever the pools price BUCK above $(1 + p^*\text{eps})$ bundles – $\text{bvib} < 1/(1 + p^*\text{eps})$ – an arbitrage exists: buy BUCK from the basket, sell it in the pools. So

$\text{bvib} \geq 1 / (1 + p^*\text{eps})$ always, up to conversion capacity,

and $\geq 1/(1+\text{eps})$ with no capacity bound at all on the bundle path. Weak side: the basket buys 1 BUCK for $(1 - \text{eps})$ bundles; whenever the pools price BUCK below $(1 - p^*\text{eps})$ bundles the arbitrage runs the other way:

$\text{bvib} \leq 1 / (1 - p^*\text{eps})$ while $R_w > 0$, up to conversion capacity.

The band is asymmetric in what it is backed by: issuance (unbounded) on the strong side; NAV on the weak side. The monetary book records issued and absorbed BUCK; the persistence test escalates a standing book to outright (Q2 burn of absorbed BUCK; Q4 acknowledgment of issued BUCK as permanent supply, K then carrying it) exactly as the desk does today.

The weak-side ladder. Let R_0 be the weak-side reserve in bundle units and $\rho = R / R_0$ the fraction remaining. Tranche k ($k = 1, 2, \dots$) bids for BUCK at $(1 - \text{eps} - (k-1) \text{ delta})$ bundles per BUCK and is sized at $\phi R_{\{k-1\}}$, where $R_{\{k-1\}}$ is the reserve remaining when it is struck. After n filled tranches

$R_n = R_0 (1 - \phi)^n$ (never zero at finite n) $b_n = 1 - \text{eps} - n \text{ delta}$ (the next bid, bundles per BUCK) $\text{bvib} \leq 1 / b_n$ (the deviation the ladder permits) $S_n = R_0 (1 - (1 - \phi)^n) / b_{\text{avg}}$ (cumulative BUCK absorbed)

so the deviation an attacker can force grows only as the LOGARITHM of the reserve fraction spent, while its average sale price worsens linearly in n . Limits: $\text{delta} = 0$ is the flat edge (finite reserve at one price – the cliff); delta/ϕ held fixed as both $\rightarrow 0$ is the constant-product bid, whose price-versus-size curve is the basket's existing full-range position; the stableswap family interpolates. Two couplings to K follow. As tranches fill, bvib rises past $1/(1 - \text{eps})$ into the sloped region – the raw error grows – AND the absorbed inventory I_d grows, raising $\text{bvib}_{\text{shadow}}$; both push K to contract, so reserve depletion is what ACCELERATES the structural correction rather than what precedes its failure. λ_d is therefore tuned with the ladder present (3c before 3d). The strong side's convenience path uses the same rule against conversion capacity: p_k rises as the epoch's C_{leg} is consumed; the bundle path is flat because it is unbounded.

The per-leg bound C_{leg} is a liveness constraint (the redemption guard must never brick), not a policy bound; it is the reason a one-day squeeze larger than the day's conversion capacity still leaves a transient trough, and the reason the directional dynamic fee (3b) and the perp's funding (3f) exist.

7.5 The attacker's cost, before and after

Today (squeeze of size S in the BUCK/USDC pool): cost = slippage against the pool's depth on the way in and out + fees, paid to the pool's LPs and the arbs; damage unbounded until the cast and K respond; at x4 undefended, damage lasts a year and the whale ends flat.

Under the ladder: the pool price is capped at $\text{par} + p^*\text{eps}$ by the undertaking, so beyond the band the whale buys from the basket; cost $\geq S * (p^*\text{eps entry} + p^*\text{eps exit}) + \text{fees} + f * \text{duration}$ (if the position is expressed or hedged in the perp). Damage: $|\text{bvib} - 1| \leq p^*\text{eps}$ outside the conversion-capacity window; inside it ($S > C_{\text{leg}}$ per epoch), the transient is bounded by what the cast and the directional fee price at tick speed. The harvest per attack is small and certain – $\sim 2 p \text{ eps } S$ – and accrues to the single receipt class; the attacker's expected value is negative at every

size, so the rational attacker does not attack. This is the target-zone result: the band converts the harvest into deterrence.

7.6 Reserve adequacy on the weak side

The weak side is the reserve-bounded flank. With a flat edge its credibility condition would be

$R_w \geq \max \text{ over plausible scenarios of } \int_{\tau_I} (\text{net BUCK sold to the basket at } \text{par} - \text{eps})$

– the reserve must outlast K’s correction of the deviation producing the outflow, and the attack is timed to the day it does not. With the ladder the condition changes character: the reserve is never exhausted at a finite price, so adequacy becomes a statement about the DEVIATION the system tolerates per fraction of reserve spent – the choice of (δ, ϕ) – and about whether K’s correction arrives before the deviation the ladder permits at the plausible cumulative outflow exceeds the tail gate. Formally: for the worst plausible outflow S over τ_I , the ladder’s $n(S)$ satisfies $1 / (1 - \text{eps} - n(S) \delta)$ within the deviation the gates allow, with the shadow-coupled K restoring ρ toward 1 inside τ_I . The catalogue’s x4 grind-down with the full cast (parity held, K 0.936, 17% railed) is the empirical frontier at today’s NAV with an implicit CPMM ladder; wave 3c re-measures it across (δ, ϕ) and wave 3h with the seeder. Sources of R_w beyond NAV: the desk’s capitalization grant and treasury accrual (existing), and the cat-bond vault (D6, optional). Nothing else.

7.7 The facility

Participant i has limit L_i , drawn d_i , thresholds (x_i, y_i) , and a deployment cap m_i . Hazard of issuing rises with the premium beyond x_i and with headroom $(L_i - d_i)$; hazard of retiring rises with the discount beyond y_i and with d_i . Aggregate utilization $u = \sum d_i / \sum L_i$ is a fast actuator’s position and enters I_c . Pro-cyclicality (4.6) is bounded by K : a premium regime raises K and headroom, so the facility can issue more into the premium – which is the correct direction – and a discount regime lowers K , which forces retirement on holders near their limit, again the correct direction. The structural risk is D8: a participant whose "insured asset" is fictitious issues free BUCK into every premium with no retirement obligation that bites (9.1).

7.8 The perp funding lever

A perpetual settled on a medianized bvib TWAP, with funding $f = g(e)$ paid continuously by the side positioned AGAINST par (longs-bvib pay when $\text{bvib} > 1$; shorts-bvib pay when $\text{bvib} < 1$) to the house, which is the monetary book. Because the band bounds bvib, the perp’s payoff is bounded, and its equilibrium funding is the market’s price for holding a band-break bet – the currency board’s automatic interest-rate mechanism applied to speculators only. Solvency condition for listing: for the defended excursion envelope measured in wave 2 (peak x duration at $x1..x4$), the funding $+ \text{eps} + \text{fee}$ surge an attacker pays to express the raid through the perp must exceed the payout the bounded excursion can return; OI is capped against the defense budget. If it does not price safely, it is not listed.

7.9 Predictions the catalogue can falsify

1. **Band integrity.** With the strong-side undertaking at $\text{eps} = 3\%$, $p = 2$, the x4 squeeze (\$40M in a day) never takes bvib below $1/(1 + p \cdot \text{eps}) \sim 0.943$ beyond the conversion-capacity window, and K ends interior (< 0.90). Today: 0.69 and 0.95 railed.

2. **Depositor fairness.** With the stress fee, depositor realized P&L in every x4 cell is $\geq$ its value in the x1 cell of the same arm. Today: $-\$1.99\text{M} / -\2.72M .
3. **Attacker cost.** Whale P&L ≤ -2 p eps S on every undertaking-bounded squeeze; the whale never ends flat on a dump (today: $\$0.00\text{M}$ at x4).
4. **Mid-ranging.** With λ_d at its knee, the desk's inventory returns to within 10% of zero inside $1.5 \tau_I$ after every x1/x2 injection, and K's rail margin at x4 is at least that of the "all" mix today ($K_{\text{end}} \leq 0.81$). With $\lambda_d = 0$ the desk's inventory grows monotonically on the drift arm (runaway one).
5. **Benign-flow health.** Quiet-regime tracking error of every basket pool against the truth-implied price is unchanged within x1.25 under the undertakings, the seeder, and the directional fee.
6. **The middle of the ladder.** A facility population scaled to the "plentiful un-leveraged asset base" cuts the x1 squeeze peak and AUC below the "credit" mix's and REALIZES (not merely marks) its harvest on the retire leg.
7. **Duration pricing.** With the perp, the whale's accumulation-phase cost rises with the days it takes, and the detector fires at least 20 days before the dump with a false-positive rate on the control below one event per two years.
8. **The ladder has no cliff.** Across the (δ, ϕ) grid on the x4 dump and grind-down, no cell shows the weak-side reserve fraction ρ below 0.1 while bvib stays inside the first edge; the whale's average sale price worsens monotonically with cumulative size in every cell; and the ladder cells reach a LOWER peak deviation per dollar of reserve spent than the flat-edge ($\delta = 0$) cell at the same eps.

Any of these failing changes a decision; 9.6 says which.

8 The wave-3 program

The catalogue's rule applies throughout: no new machinery before the falsifiable instrument, and no contract work before a catalogue win. Cells are `catalogue.py` grid entries (arm x mix x scale, plus `--set` overrides); new physics arrives as agents first and contract features second, because the sim runs the real contracts and an agent with the right authority (SimLP's zero-premium credit and V3 callback proxy) can stand in for a contract feature at the economics level before the Solidity exists.

8.1 Instrument: the markout / LVR ledger, with the basket-side columns (3a)

Definitions. For each swap s in constituent pool i at time t , with the basket-LP side's signed base-token flow q_{lp} and execution price p_{exec} :

$$\text{markout}_D(s) = q_{lp} \times (p_{ref}(t + D) - p_{exec}(t))$$

Reference for the COMMON-MODE ledger: PAR (reversion to 1 at horizon D). For the DIFFERENTIAL ledger: the truth-implied TOKEN/BUCK, and (sim-only) the exogenous reference series. Horizons D in $\{\text{next tick}, 1 \text{ d}, 5 \text{ d}\}$. Per pool per window: fees_i , adverse_i (sum of markouts), $\text{netlp}_i = \text{fees}_i + \text{adverse}_i$, lvr_bench_i (differential only), $\text{carry_ratio}_i = \text{fees}_i / \max(\text{lvr_bench}_i, -\text{adverse}_i)$.

Attribution. Every sim swap has a known sender class: whale (accumulate / dump / re-buy separately), BUCK-unaware arbs, director keeper, excursion agents by base, DM depositors, debtors, and – new – the undertakings, the facility, the seeder, the house. Toxicity of a class = mean markout per dollar of its flow. Expected: whale dump-day flow deeply negative for the basket; benign arbs mildly negative gross, ~zero net of fee; the keeper netting to ~fees; excursion agents negative for the pool ledger by design (the defense partly funded by the basket’s own book – confirmed in outline by the x1 dump: defenders +\$0.26M, basket NAV -\$0.75M).

Where it lives. `snapshot.py`: swap-event capture with class tags and deferred markout at $t + D$; `basketNav` / `treasuryBuck` / `dmProfitUsd` promoted into the catalogue summary. `eqmetrics.py`: a markout panel beside `excursion_windows` / `excursion_response`. `catalogue.py`: grid columns – basket fees, adverse, netlp, carry_ratio (worst pool), NAV USD and NAV-in-baskets delta vs undefended, treasury delta, depositor realized delta, whale toxicity.

Cells. The existing 86 cells, re-summarized from their vectors (`--report-only` after the metric lands); no new physics.

Gate. None new for 3a; it is the instrument. Deliverable: the table in 3.3 produced by `catalogue.py` rather than by hand.

8.2 Gates for every later wave

1. The 2-year tail gate (`eqmetrics.accept`): tail bvib within 1.00 +/- 0.02, K never railed in the tail, channels alive.
2. **Carry coverage**: tail carry_ratio_i >= 1 for every constituent pool.
3. **Benign-flow health**: quiet-regime tracking error of each pool vs the truth-implied price within baseline x1.25.
4. **Duress-exit fairness**: depositor realized P&L in a defended or undertaking cell >= the undefended cell of the same arm; the transfer between exiting and remaining depositors has the remaining side positive.
5. **Band integrity** (from 3c on): bvib within $[1/(1+p*\epsilon), 1/(1-p*\epsilon)]$ at every frame outside the conversion-capacity window of an injection; the undertakings’ books return to zero within 1.5 tau_I.
6. **Attacker cost**: whale P&L <= -2 p eps S on bounded squeezes; never flat on a dump.

8.3 The waves

wave	axis	cells
3a	instrument	existing 86 cells, re-summarized
3c	the undertakings	eps {2,3,5}% x p {1.5,2,3} x {dump, squeeze, grind-down, grind-up} x {x1, x4}; undefended AND full cast; w
3d	shadow-bvib controller	lambda_d {0, 0.25, 0.5, 1, 2} with <code>--basket ops</code> ENABLED in the loop (first time), on dump/squeeze x1/x4
3b	directional dynamic fee	{static-retune, directional-RV, directional-size} x {dump, squeeze} x {x1, x4}
3g	the facility	population {8, 32, 128} x (x, y) {(2%,2%), (3%,3%), (5%,2%)} x deposit-to-basket on/off, on squeeze/dump
3f	perp funding lever	g() slope {0, 1, 3, 10} x whale-via-perp vs whale-via-spot, on squeeze x1/x4 and the accumulation phase of d
3e	derivatives on satellites	D1 hedge on/off x {spike, step} x {x1.3, x1.6}
3h	seeder, seeds, floods	seeder on/off at launch; the campaign’s 5 seeds on the key 3c/3d cells; the oversupply flood (rec. 4) with and

Order: 3a -> 3c -> 3d -> 3b -> 3g -> 3f -> 3e/3h. 3c precedes 3d because the undertakings change the desk’s book size, which is what the lambda sweep must be measured against; 3g precedes 3f because the middle of the ladder decides how much duration pricing is left to do.

8.4 Implementation notes for the sim

- **UndertakingAgent** is the fastest honest stand-in for D3. SimLP can mint against a zero-premium credit with an arbitrarily large face and hold both sides of a V3 position; the basket-rotation execution path already trades TOKEN $\leftrightarrow$ BUCK in every constituent pool in `basketAmount` proportions. Standing orders at par $\pm$ eps in bundle units are then eight range orders (one per pool) re-struck daily, or, more simply, a daily check that fires when `bvib` crosses the edge and trades up to `C_leg` per pool. The agent must book its issued and absorbed BUCK so 3d can read `I_d`.
- **The shadow controller** needs Solidity: `BuckKControllerDirect` reads `basket.basketValueInBuck()`; a `Shadow` variant reading a new `shadowValueInBuck()` on `BuckBasketOps` (`bvib + lambda x monetaryDeviationOffset() / depth`) is small, and the sim deploys whatever the repo compiles. Until the undertakings are contract-level, the agent's booked inventory must be pushed into the offset (a governance setter on the ops shell for the sim, removed later).
- **The facility** is the credit base with two changes: the issue leg deposits into the basket instead of selling into BUCK/USDC, and the retire leg redeems the receipt and repays. Utilization telemetry is a counter in `snapshot.py` (add it in the same commit – the ops doc's "the agent is invisible" lesson).
- **The perp** is a Python-side ledger (positions, mark = medianized `bvib`, funding accrual) whose funding settles on-chain as BUCK transfers between the whale's proxy and the house's proxy each tick; the whale gains a `via = "perp"` knob. No contract until 3f wins.
- **The treasury seeder** is `BuckPoolInvestorAgent` with the range struck from the current BUCK/USDC tick toward the truth-implied ideal, a withdraw rule keyed to SimLP-independent depth, and the ops doc's orientation checklist applied.
- **The directional fee** is a fee schedule in the pool adapter used by the router quoter and the sim's swap path; the v4 hook comes later behind the venue seam of `BASKET-REDESIGN.md` 4.
- **Telemetry discipline** (ops doc): counters into `snapshot.py` in the same commit as the agent; per-agent inventory through `arb_state`; several seeds before any peak-damping number is quoted.

8.5 Compute

At ~ 1 core-hour per 2-year pyrevm cell (the catalogue's measured rate), 3a is free, 3c is ~ 72 cells, 3d ~ 40 , 3b ~ 24 , 3g ~ 72 , 3f ~ 24 , 3e ~ 8 , 3h ~ 40 – ~ 280 cells, a few days of a MATRIX_JOBS-way machine, in the order above so each wave can stop the next.

9 Risks, open questions, and what would change the decisions

9.1 D8 and the facility

The facility lets latent credit issue into premiums. A participant whose insured asset is fictitious (D8: `createCredit` is permissionless and zero-premium credit is exempt from the funding gate) issues free BUCK into every premium with a retirement obligation that never bites (no real asset, no reason to close). This is the oversupply flood of [REBALANCING-EQ.org](https://rebalancing-eq.org) recommendation 4, automated. The facility must be restricted to vetted insurers' credit (the on-chain solvency-plus-rating scheme the

master doc names as the mainnet blocker) or, until then, to a whitelisted insurer set in sim and pilot. 3h measures the flood with and without the facility to size the exposure.

9.2 Band-break credibility on the weak side

The weak side is NAV-bounded. Bertola-Caballero says an imperfectly credible edge destabilizes near it, and Krugman-Flood-Garber say the attack is timed to the day a flat edge stops being honored. The ladder (D3 refinement, 7.4) is the structural mitigation: there is no day on which the weak side stops being honored, only successively poorer prices, so there is no discontinuity to time and no size to probe – the RULE (the tranche schedule against the on-chain reserve) is what is announced. The residual risk moves to the ladder’s parameters: a delta too small reproduces the cliff, a delta too large lets a modest outflow force a deep deviation. Wave 3c’s (delta, phi) grid, the perp pricing band-break bets, and a reserve visibly large relative to tau_I outflow (7.5) are the answers.

9.3 Sensor damping by the invisible cast

Private defenders damp bvib and K cannot see their books. Their capital is finite and their positions temporary by design, so the lag is bounded by their size; but a very large private cast makes K late. The shadow term could include an estimate of private inventory from flow attribution; it is deliberately left out until 3d shows the desk-only term is insufficient.

9.4 Conversion capacity and the one-day trough

The bundle path is unlimited but the convenience path and every constituent-pool leg are bounded by C_leg for liveness. A one-day squeeze larger than a day’s capacity still troughs. The directional fee is the tick-speed answer; if 3b shows it insufficient, the remaining lever is a larger per-leg bound with a redemption guard that tolerates it (a TWAP-window-aware guard rather than a flat maxLegBp).

9.5 Governance abandonment

The SNB failure mode is available to any governance that can move par. Par is definitional here and the undertakings are rules, not operations; the residual is a governance that changes eps, p, lambda or the basket under pressure. Time-locks on those parameters, and the announcement discipline of Hong Kong’s technical measures, are the answer; they are policy, not code.

9.6 What would change which decision

observation in wave 3	changes	to
band integrity fails at x4 even with unlimited strong side	D3 (eps, p)	larger eps, larger C_leg with a tolerant guard,
the ladder’s peak deviation per reserve dollar is no better than the flat edge	D3 (delta, phi)	steeper or shallower schedule, or the CPMM li
depositor realized P&L still negative with the stress fee	D2	a larger fee schedule, or a redemption-reserve t
desk inventory never returns to zero at any lambda	D4	K’s tau_I, or an explicit Q2 rule keyed to inve
the facility issues into premiums but never retires	D6	mandatory retire thresholds, or facility restrict
the perp raises attacker cost less than its liquidation risk costs the house	D5	do not build the perp; rely on the band and th
benign tracking error rises under the seeder or the fee	D5, 3b	narrower seeder range; lower base fee
the flood equilibrates below theta x apr with the facility on	D6 + D8	the facility waits for insurer vetting

9.7 The JS port

`alberta-buck-platform.org` ports only agents whose excursion control the catalogue demonstrated, in their simplest embodiments. The undertakings, the shadow controller and the stress fee are CONTRACT features and port with the contracts; the facility, the seeder and the house are agents and port only after their cells win. The perp ledger is Python-side in 3f and is not a port candidate until it is a contract.

10 Recommendations, ranked by expected information per unit of compute

1. **Build 3a**, including the basket-side columns – the table in 3.3 belongs in the summary, not in a scratchpad script.
2. **3c in the sim** – the undertakings are the largest single change to the physics and the x4 squeeze is the cell that decides them.
3. **3d** – put the desk in the loop for the first time, with the shadow term; find lambda’s knee the way the 20-day knee was found.
4. **3b** – the directional fee, for the one-day trough.
5. **3g then 3f** – the middle of the ladder, then the duration lever.
6. **3e, 3h** – satellites, seeds, the seeder, the flood.
7. **Contract work LAST**, in this order: the undertakings and monetary-book accounting on the ops shell; `shadowValueInBuck` and the `Shadow` controller; the stress fee in `_redeem`; the facility; a v4 hook for the winning fee policy behind the venue seam; the perp only if 3f closes. Nothing ships on-chain that did not first win its catalogue cell.

11 Glossary

- **bvib**: `basketValueInBuck`, the BUCK cost of one basket bundle; the controller’s process variable; > 1 = BUCK cheap (discount), < 1 = BUCK dear (premium).
- **common mode / differential mode**: BUCK against the bundle (K’s job) vs constituent against constituent (the director’s job); the mean and the differences of the same tick-EMA ladders.
- **the desk**: `BuckBasketOps`; Q1 absorb, Q2 retire, Q3 supply, Q4 issue.
- **the undertakings**: the strong-side (sell BUCK, unlimited) and weak-side (buy BUCK, NAV-bounded) standing offers at par +/- eps in bundle units; the currency board.
- **the weak-side ladder**: the weak side struck as tranches at successively deeper discounts ($\text{eps} + (k-1) \text{delta}$), each sized at a fraction ϕ of the reserve remaining; no cliff, superlinear attacker cost, and a deviation that gauges reserve stress.
- **the facility**: the credit auto-pay-down facility; latent credit issuing into premiums and retiring into discounts on user-set thresholds.

- **the shadow bvib:** bvib plus the inventory the visible fast actuators have absorbed minus issued, scaled; what K sees.
- **the seeder:** the treasury's temporary BUCK/USDC position from the current price toward the ideal.
- **the house:** the monetary book as counterparty and funding recipient of the bvib perp.
- **the cast:** the private excursion agents of the catalogue (usdc, buck, credit, basket, crb).
- **manufactured reversion:** the property that par is definitional and K plus the cast force bvib back to it – what makes selling common-mode convexity sound.

12 References

12.1 Repository documents and code

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